

Algebra II with Trigonometry

Chapter 5.4

Olympiad Problems

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[4401270_alg-2trig-h-chp-5-4-note-docx.pdf](#)

Problems

1. Compute the exact value of $\arccos\left(\sin\left(\arctan\frac{3}{4}\right)\right)$.
2. Given that $\theta = \arcsin\left(\frac{5}{13}\right)$, find $\cot(2\theta)$.
3. If $\arctan x + \arctan y = \frac{\pi}{4}$, what is the value of $x + y + xy$?
4. Let $a = \arccos\left(\frac{1}{3}\right)$ and $b = \arcsin\left(\frac{2\sqrt{2}}{3}\right)$. Without a calculator, determine whether $a = b$.
5. Evaluate $\sin\left(\arccos\left(\frac{x}{\sqrt{1+x^2}}\right)\right)$ in terms of x .
6. Find all real x such that $\arcsin x + \arccos x = \arctan(2x)$.